CIS 215 LAB REPORT

Network Security Fundamentals

6.6.4 Packet Tracer — Configure and Verify NTP

BACKGROUND	
Course:	CIS 215 - Network Security Fundamentals
Term:	Spring 2026
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Report Title:	6.6.4 Packet Tracer - Configure and Verify NTP
Week / Module Reference:	Week 6 / Time Synchronization and Management Security
Assignment Reference:	Lab 6.6.4 / Packet Tracer Activity
Lab Type:	Hands-On Lab / Packet Tracer
Date Completed:	2026-04-21
Time Spent:	Approximately 1.5 hours

ENVIRONMENT

Platforms, Devices, and Systems Used:

- Primary host system: Windows desktop environment with Cisco Packet Tracer used for build, configuration, testing, and evidence capture.
- Lab platform: Cisco Packet Tracer simulation with a small NTP topology built manually from a blank workspace.
- Devices used: N1 server, routers R1 and R2, and switch S1.
- Router platform used in the simulation: ISR4321; switch platform used: 2960-24TT; server platform used: Server-PT.
- Security focus area: secure time synchronization using authenticated NTP between an NTP server and client routers.

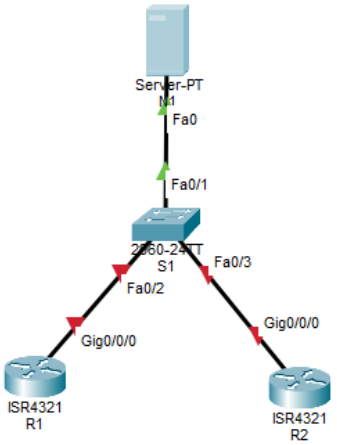

Figure 1. Initial topology state after the server, switch, and routers were placed and cabled, before the router interfaces were brought up.

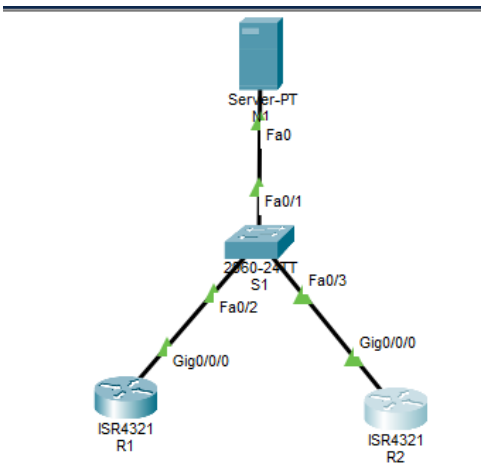

Figure 2. Topology after the router interfaces were enabled and the physical connections transitioned to an operational state.

Files, Media, and Resources Used:

- Packet Tracer source file: completed 6.6.4 Configure and Verify NTP .pkt lab file.
- Lab instructions: assignment sheet containing the addressing table, NTP server/client requirements, and verification commands.
- Evidence collected: topology build screenshots, interface bring-up screenshots, failed and successful connectivity verification, unsynchronized NTP status output, synchronized NTP verification, and final topology screenshots.
- Supporting notes: addressing checklist, interface map, authentication key details, and final verification command output.

Work Objective:

- Build a working local topology from a blank Packet Tracer workspace using one server, two routers, and one switch.
- Configure IP addressing so both routers can reach the NTP server on the shared subnet.
- Configure R1 and R2 as authenticated NTP clients of N1, then verify synchronized clocks using the required show commands.

Learning Objectives:

- Interpret a small addressing table and map it to a blank Packet Tracer build.
- Configure router interfaces and a server NIC on a single IPv4 subnet.
- Verify basic reachability with ICMP before attempting management-plane configuration.
- Configure NTP server association, NTP authentication, and trusted keys on Cisco routers.
- Use show ntp status, show ntp associations, and show clock detail to verify full synchronization.

Scope and Constraints:

- In scope: topology build, interface bring-up, IP addressing, server configuration, ping verification, NTP client setup, NTP authentication, synchronization validation, and final save.
- The work was completed entirely inside Cisco Packet Tracer using manual device placement, manual cabling, and CLI-based router configuration.
- The simulation used ISR4321 routers, which remained acceptable because the required interface identifiers and NTP commands were available and the intended lab outcomes were preserved.

Addressing Plan Used:

- N1 NIC: 209.165.200.225 / 255.255.255.0
- R1 G0/0/0: 209.165.200.226 / 255.255.255.0
- R2 G0/0/0: 209.165.200.227 / 255.255.255.0
- All devices shared the same 209.165.200.0 /24 network through switch S1.
- NTP authentication details: key 1, MD5 password NTPpa55.

PART 1 — TOPOLOGY BUILD AND BASIC CONNECTIVITY

Preparation and Physical Build Procedure:

1) Placed one Server-PT device and renamed it N1.
2) Placed two ISR4321 routers and renamed them R1 and R2.
3) Placed one 2960-24TT switch and used it as the shared Layer 2 segment for the local subnet.
4) Connected N1 FastEthernet0 to S1 FastEthernet0/1 with a copper straight-through cable.
5) Connected R1 GigabitEthernet0/0/0 to S1 FastEthernet0/2 with a copper straight-through cable.
6) Connected R2 GigabitEthernet0/0/0 to S1 FastEthernet0/3 with a copper straight-through cable.
7) Confirmed that the topology matched the addressing table and that all three devices shared one common Ethernet segment.

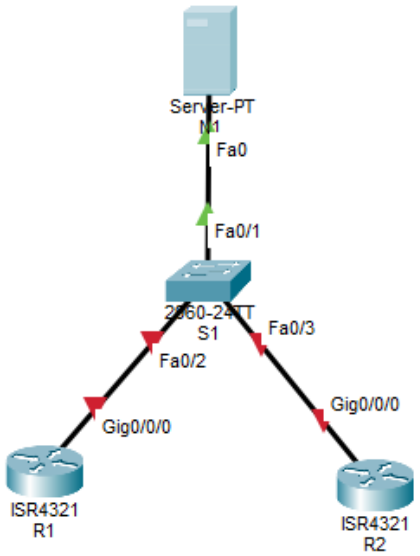

Figure 3. Initial build state after device placement and cabling, before the router interfaces were enabled.

Router Bring-Up and Basic Interface Configuration:

1) Entered privileged EXEC mode on R1 and R2 and skipped the initial configuration dialog.
2) Applied hostname R1 on the first router and hostname R2 on the second router.
3) Entered interface g0/0/0 on each router and issued no shutdown to activate the physical links.
4) Observed the link-change messages confirming that the interfaces transitioned to an operational state.
5) Configured the IPv4 address 209.165.200.226 /24 on R1 g0/0/0.
6) Configured the IPv4 address 209.165.200.227 /24 on R2 g0/0/0.
7) Verified the configured interfaces with show ip interface brief before moving to server validation.

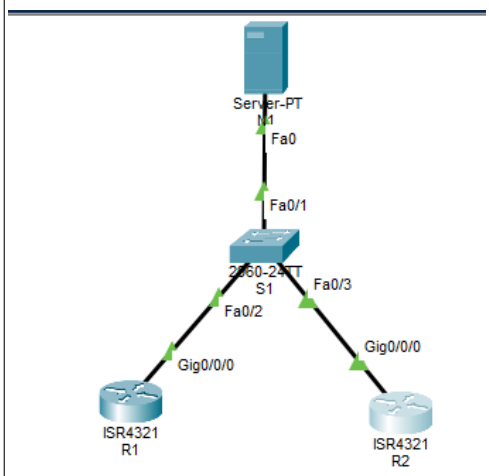

Figure 4. Operational topology after no shutdown was applied and the switch-facing router links came up.

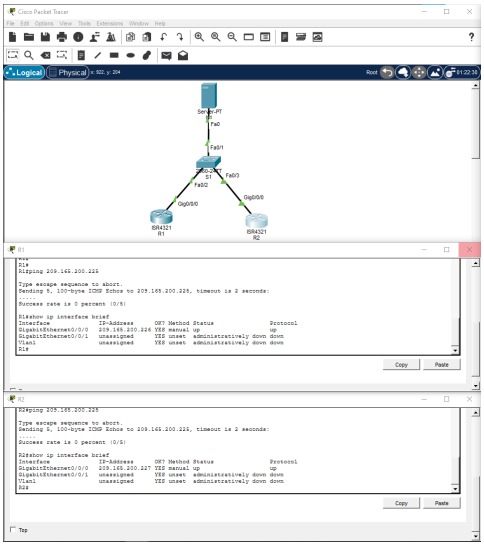

Figure 5. Early connectivity validation showing correct router interface addressing but failed pings before the server-side setup was fully complete.

Additional Notes for Part 1:

- The switch was not explicitly listed in the addressing table, but it was required in practice because all documented devices shared the same /24 Ethernet network and needed a common Layer 2 segment.
- The failed ping stage was useful evidence because it documented the troubleshooting sequence rather than hiding the intermediate state.
- The interface naming on the ISR4321 routers matched the lab requirements, so the Packet Tracer build remained aligned with the task.

PART 1 — CONTINUED

Server Configuration and Connectivity Validation:

1) Configured N1 with IPv4 address 209.165.200.225 and subnet mask 255.255.255.0.
2) Verified the server remained connected to S1 through FastEthernet0.
3) Tested reachability from R1 to 209.165.200.225 using ping.
4) Tested reachability from R2 to 209.165.200.225 using ping.
5) Observed an initial timeout due to ARP resolution, followed by successful replies represented by the .!!!! pattern.
6) Used the successful ping results as proof that the addressing and local Layer 2 connectivity were correct before configuring NTP.

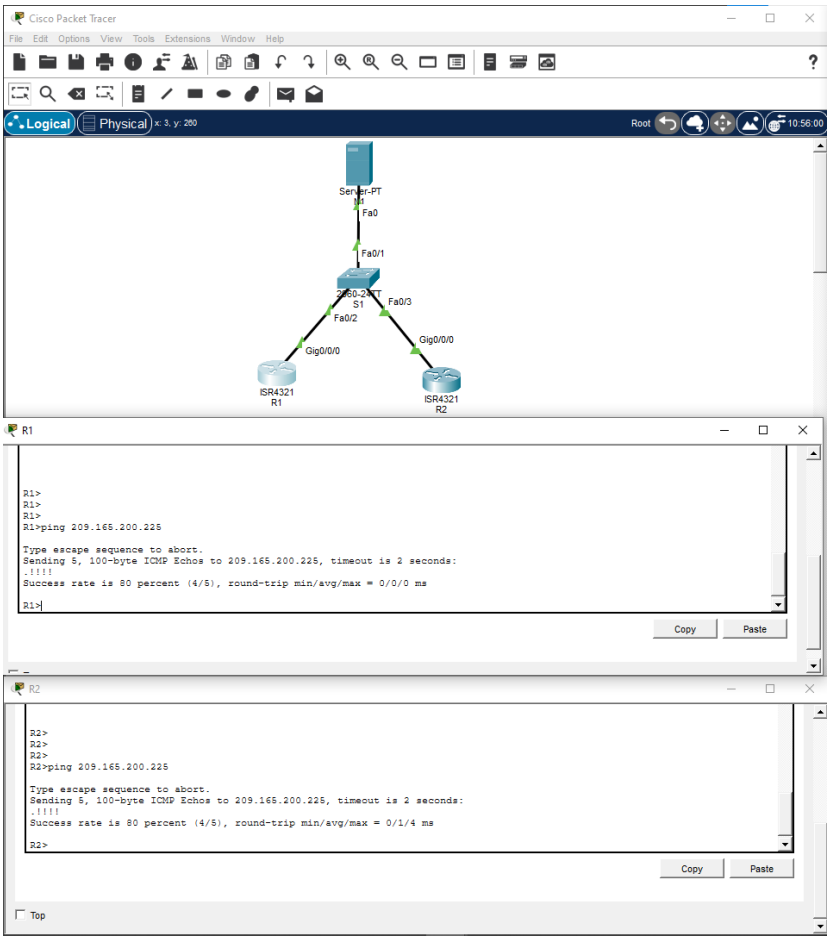

Figure 6. Successful connectivity verification between both routers and the NTP server after IP addressing and ARP resolution completed.

Key Part 1 Commands:

BASIC SETUP	ADDRESSING	SERVER / VERIFY
enable	ip address 209.165.200.226	IP Address: 209.165.200.225
configure terminal	255.255.255.0	Subnet Mask: 255.255.255.0
hostname R1	ip address 209.165.200.227	show ip interface brief
hostname R2	255.255.255.0	ping 209.165.200.225
interface g0/0/0	show ip interface brief	copy running-config startup-config
no shutdown	ping 209.165.200.225	
show ip interface brief		
show running-config		

Part 1 Result:

- Part 1 was completed successfully because the topology was built correctly, router interfaces were active, the server was addressed properly, and both routers successfully reached the NTP server through ICMP testing.

PART 2 — NTP CONFIGURATION AND VERIFICATION

NTP Client Configuration Procedure:

1) Verified N1 under Services > NTP and confirmed it was acting as the NTP server for the topology.
2) Confirmed that NTP authentication on N1 used key 1 and password NTPpa55.
3) Checked the initial status on the routers and observed that NTP was not yet synchronized.
4) Configured ntp server 209.165.200.225 on R1 and R2.
5) Enabled ntp authenticate on both routers.
6) Configured ntp trusted-key 1 on both routers.
7) Configured ntp authentication-key 1 md5 NTPpa55 on both routers.
8) Allowed synchronization time to pass and used Fast Forward Time in Packet Tracer to accelerate convergence.

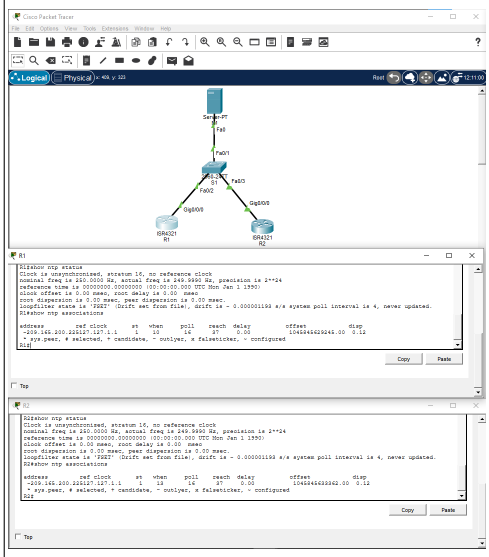

Figure 7. NTP configured but not yet synchronized, showing the intermediate unsynchronized state before authentication and timing fully converged.

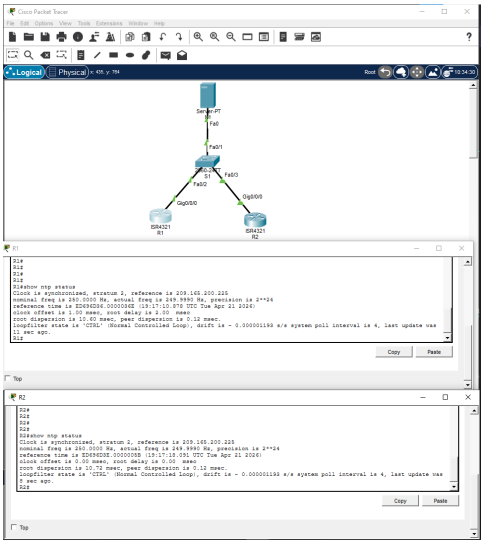

Figure 8. Final NTP synchronization verification showing both routers synchronized to the server at 209.165.200.225.

Verification and Result Interpretation:

- show ntp status initially reported Clock is unsynchronized, stratum 16, which confirmed that configuration alone does not immediately guarantee synchronization.
- show ntp associations initially displayed the server as configured with but not yet selected with *, which indicated an intermediate state rather than final synchronization.
- After the authentication details were confirmed on N1 and sufficient time passed, show ntp status reported Clock is synchronized, stratum 2, reference is 209.165.200.225 on both routers.
- show ntp associations then displayed * 209.165.200.225, confirming that the configured server became the active synchronized peer.
- show clock detail reported Time source is NTP, which verified that the routers had moved from hardware time to the network time source.

Key Part 2 Commands:

PRE-CHECKS	NTP CONFIG	VERIFY
show ntp status	configure terminal	show ntp status
show clock detail	ntp server 209.165.200.225	show ntp associations
ping 209.165.200.225	ntp authenticate	show clock detail
	ntp trusted-key 1	copy running-config startup-config
	ntp authentication-key 1 md5 NTPpa55	
	end	

Question Answered During Verification:

1) Are the clocks synchronized? - Yes. Both R1 and R2 were synchronized using NTP, with 209.165.200.225 as the reference time source.

Part 2 Result:

- Part 2 was completed successfully because both routers were configured as authenticated NTP clients, the server at 209.165.200.225 became the active synchronized peer, and the clock source on each router changed to NTP.

SUMMARY, RESULTS, AND REFLECTION

Completion Verification:

- The topology was built manually from a blank Packet Tracer workspace and all required server, switch, and router connections were completed correctly.
- Router addressing and server addressing were applied correctly on the shared subnet.
- Connectivity between both routers and the NTP server was verified through successful ping testing.
- NTP authentication and trusted-key configuration were applied successfully on R1 and R2.
- Both routers synchronized to N1 and reported Time source is NTP.
- The final Packet Tracer file was saved in a working, submission-ready state after all required validation steps were complete.

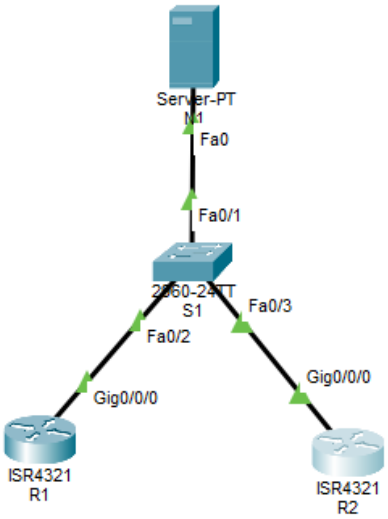

Figure 9. Final completed topology after successful connectivity verification and authenticated NTP synchronization.

Key Findings:

1) Time synchronization depends on both reachability and correct authentication, so successful pings alone do not complete the task.
2) The intermediate unsynchronized state is a useful troubleshooting checkpoint because it shows that the routers have the server configured but have not yet fully selected it as the active time source.
3) show ntp status, show ntp associations, and show clock detail complement each other and should be used together for solid verification.
4) The first successful ping sequence often includes one dropped packet because ARP resolution must occur before full reachability is established.

Security Concepts Reinforced:

- Accurate and consistent timestamps are important to event correlation, logging, incident investigation, and security monitoring.
- NTP authentication reduces the risk of trusting an invalid or malicious time source.
- Management-plane services such as NTP should be verified carefully because configuration errors can silently affect later security analysis.
- Verification should include both intermediate and final states so troubleshooting evidence is preserved.

Best Practices / Operational Notes:

- Complete the topology and basic IP connectivity first, then configure NTP.
- Confirm the authentication key and password on the server before assuming the router configuration is the problem.
- Use Fast Forward Time in Packet Tracer when needed so synchronization can occur without unnecessary waiting.
- Save the configuration after validation so the final state remains available for later review and submission.

Reflection:

- This lab reinforced that even a small management-service topology benefits from a phased workflow: build, address, verify reachability, configure the service, then verify the final operational state.
- The most useful takeaway was seeing the difference between a router that merely has an NTP server configured and a router that is actually synchronized to that server.
- From a professional standpoint, the completed work demonstrates the ability to interpret sparse instructions, build a clean topology from scratch, troubleshoot intermediate failures, and verify an authenticated management service end to end.

Artifacts and Deliverables:

- Completed Packet Tracer .pkt file: 6.6.4-configure-verify-ntp.pkt
- Supporting screenshots for build state, interface bring-up, ping verification, unsynchronized NTP state, synchronized NTP state, and final completed topology.
- Final written report documenting the topology, configuration workflow, verification, and outcomes.

Next Steps:

- Attach the completed Packet Tracer file and submit it through Canvas as required by the assignment.
- Store the final screenshots and report in the course repository as polished evidence of the completed lab.
- Reuse the same phased workflow in future labs involving logging, management protocols, secure administration, and broader management-plane security controls.